

CXR Vault Lab Manual

SW.17 — dev secrets + full wiring guide

CXR Ops Lab — `cxr-ops-lab`

2026-05-31

Abstract

SW.17: HashiCorp Vault dev mode (:8200) stores CXR-shaped secrets at **secret/cxr/***. This manual documents **exactly how it is wired** — Compose → seed script → KV → UI — and where it would attach to Claim Studio / K8 in a future phase (not live today).

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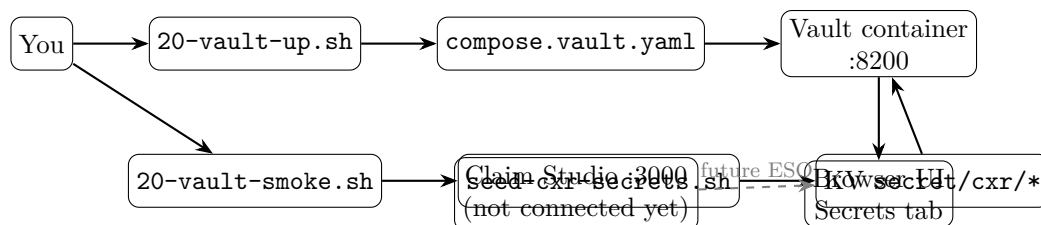
Chapter 1

How this lab is wired (step by step)

1.1 The chain (what runs when you type the commands)

1. You run `./scripts/20-vault-up.sh`
2. Script calls `docker compose -f compose.vault.yaml up -d`
3. **compose.vault.yaml** starts container `cxr-ops-lab-vault-1`:
 - Image `hashicorp/vault:1.17`, port **8200:8200**
 - Dev mode: in-memory store, auto-unsealed
 - Root token fixed: `cxr-bootcamp-root` via `VAULT_DEV_ROOT_TOKEN_ID`
4. You run `./scripts/20-vault-smoke.sh`:
 - `curl :8200/v1/sys/health` — Vault is up
 - Calls `lab/vault/seed-cxr-secrets.sh`
5. **seed-cxr-secrets.sh** `docker execs` into the Vault container:
 - `vault secrets enable -path=secret kv-v2` (KV version 2 engine)
 - `vault kv put secret/cxr/analyzer ...` (and `otel`, `datastores`)
6. You open `http://localhost:8200`, login with token `cxr-bootcamp-root`
7. UI path: Secrets → `secret/` → `cxr/` → `analyzer`

1.2 Wiring diagram



Chapter 2

What is wired vs not wired

Layer	Status today	Future (sketch)
Vault container :8200 secret/cxr/* keys Claim Studio :3000	Running via Compose Seeded by script Not connected	Production Vault cluster Same paths, stricter ACLs analyze/route.ts reads env
Rehearsal :8251	Not connected	Same
K8 pod :8081	Not connected	External Secrets → Vault → pod env

Today Claim Studio still gets `CXR_ANALYZER_SCRIPT` from `.env.local` / compose env.
This lab proves the *secret store* and *path naming* match CXR env inventory (`lab/vault/cxr-secret-map.json`)

Chapter 3

Secret map (env $\rightarrow$ Vault path)

Vault path	Key	CXR use
secret/cxr/analyzer	CXR_ANALYZER_SCRIPT	Python path for analyze
	CXR_JUDGE_SCRIPT	Judge script path
secret/cxr/otel	OTEL_*	SW.11 trace export
secret/cxr/datastores	QDRANT_URL	Vector DB
	DATABASE_URL	SQL (placeholder)

Chapter 4

UI navigation (after token login)

- **Ignore for bootcamp:** Dashboard widgets, Access, Policies, Agents
- **Go to:** Secrets → engine `secret/` → folder `cxr/`
- **Golden path:** open analyzer — confirm `/analyzers/analyze_sample.py`
- If `cxr/` is empty: re-run `./scripts/20-vault-smoke.sh`

Chapter 5

Files in the repo

File	Role in the wiring
<code>compose.vault.yaml</code>	Defines Vault service, port, dev token
<code>scripts/20-vault-up.sh</code>	Wrapper: <code>compose up -d</code>
<code>scripts/20-vault-smoke.sh</code>	Health + calls seed + reads golden secret
<code>lab/vault/seed-cxr-secrets.sh</code>	<code>docker exec vault kv put ...</code>
<code>lab/vault/cxr-secret-map.json</code>	Documentation of paths (source of truth for names)
<code>evidence/SW17-vault-verify-2016-05-31.md</code>	

Chapter 6

Hypothetical K8 wiring (not deployed)

```
# Future sketch only:
# 1. External Secrets Operator installed on kind cluster
# 2. ExternalSecret CR references secret/cxr/analyzer
# 3. Helm chart cxr-ui deployment envFrom -> synced K8 Secret
# 4. Pod at :8081 gets CXR_ANALYZER_SCRIPT without git/env files
```

Chapter 7

Live adjustments

- **Dev token in compose** — never use `cxr-bootcamp-root` in production.
- **KV enable idempotent** — seed script ignores “already enabled”.
- **Smoke retry** — Vault needs a few seconds after up; smoke waits on health.
- **Token login only** — no username/password in dev mode.

Chapter 8

Procedure

```
cd cxr-ops-lab
./scripts/20-vault-up.sh
./scripts/20-vault-smoke.sh
# Browser: http://localhost:8200   Token: cxr-bootcamp-root
# Secrets -> secret/cxr/analyzer
```

Chapter 9

Stop

```
docker compose -f compose.vault.yaml down
```